1. Perceptron (artificial neuron):

A. perceptron is type of artificial neuron.

B. Input: $x_i \{ i \mid 1,2 ...... n \}$

C. Bias Input: $x_0 = 1$ (always)

D. Weight: $w_i \{ i \mid 0,1 ... n \}$

E. Computational Unit: $\boxed{z = \sum_{0}^{n} w_i \cdot x_i}$ (weighted sum function) + $\boxed{f(z) = \begin{cases} -1, & z < 0 \\ 1, & z \geq 0 \end{cases}}$ (activation function)

F. Output: $y = f(z)$

G. sign function is a specific activation function used by perceptron

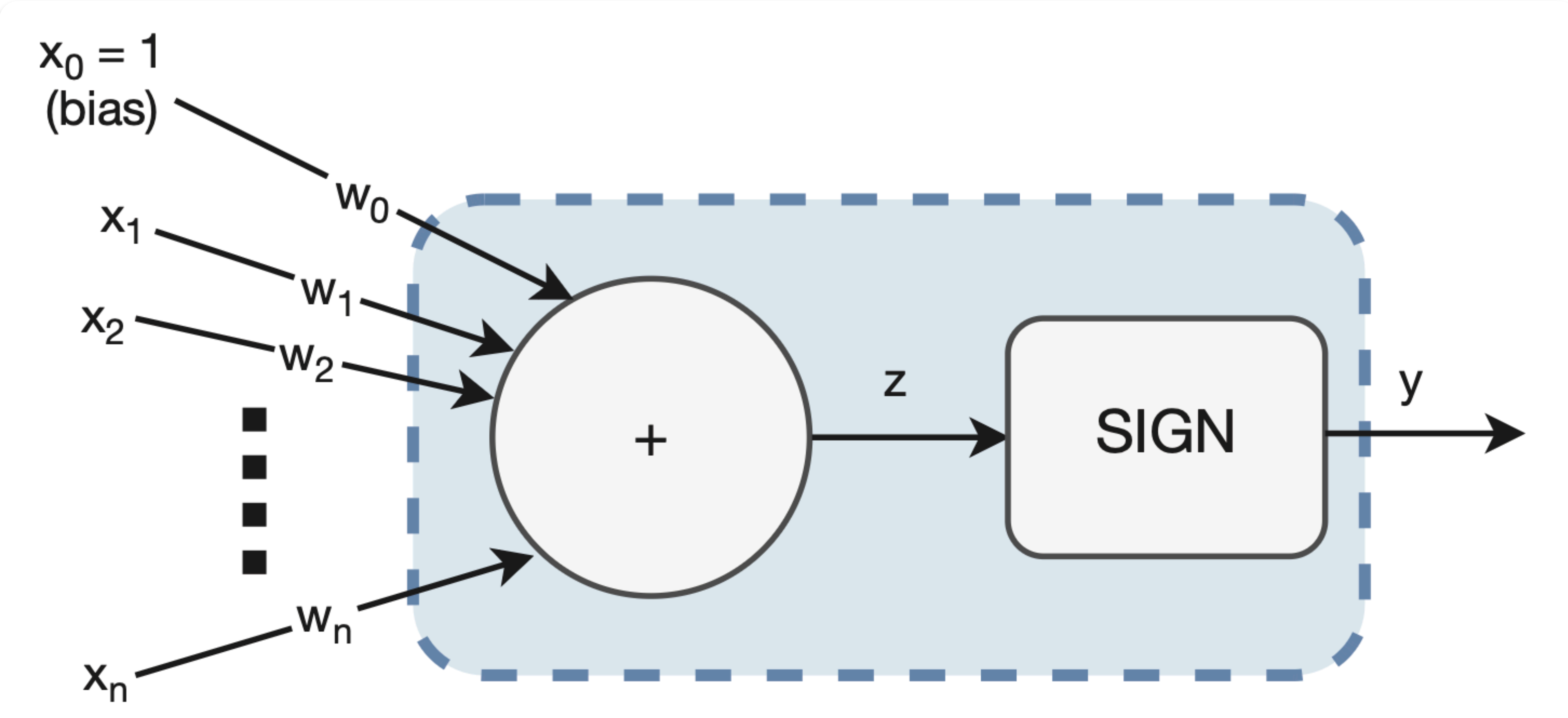
H. 

Figure 1-2 The perceptron

2. Neural Network (combined with few artificial neuron):

A. input & output of a perceptron can be " a range of real number.

B. each perceptron can implement more complex function than NAND gate, AND gate due to A.

C. use " learning algorith" & "observed example", weight of each perceptron can be adjust to design a more accurate neural network automatically.

D. the rebuilt neuron network tends to generalize that can make sense for " not-yet-observed example" and get correct ouputs.